

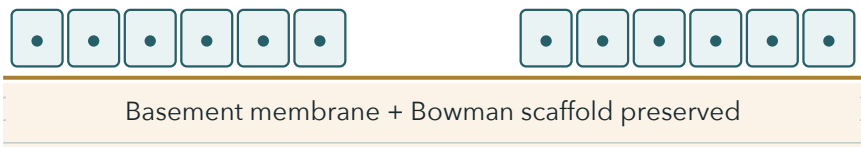
# Corneal barrier and healing

Depth of injury predicts the repair route and the risk of an optical scar.

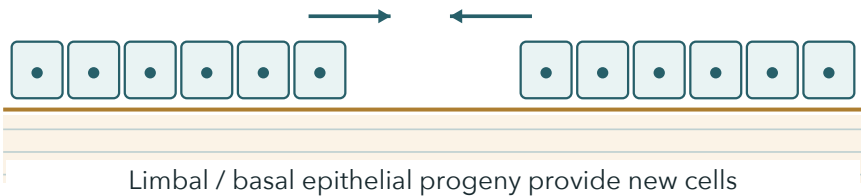
## A superficial barrier can renew

### Closing a surface defect

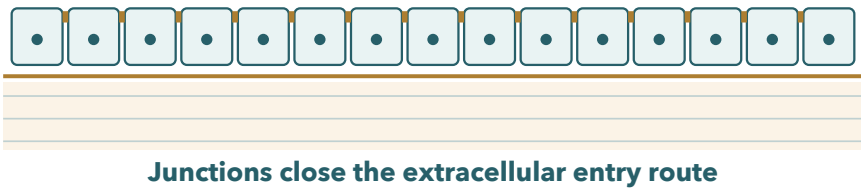
#### 1 SURFACE DEFECT



#### 2 CELLS MIGRATE, THEN REPLENISH



#### 3 JOINED SURFACE BARRIER



Superficial abrasion with preserved underlying scaffold. The sequence is simplified; migration and proliferation can overlap.

### Barrier, attachment, replacement

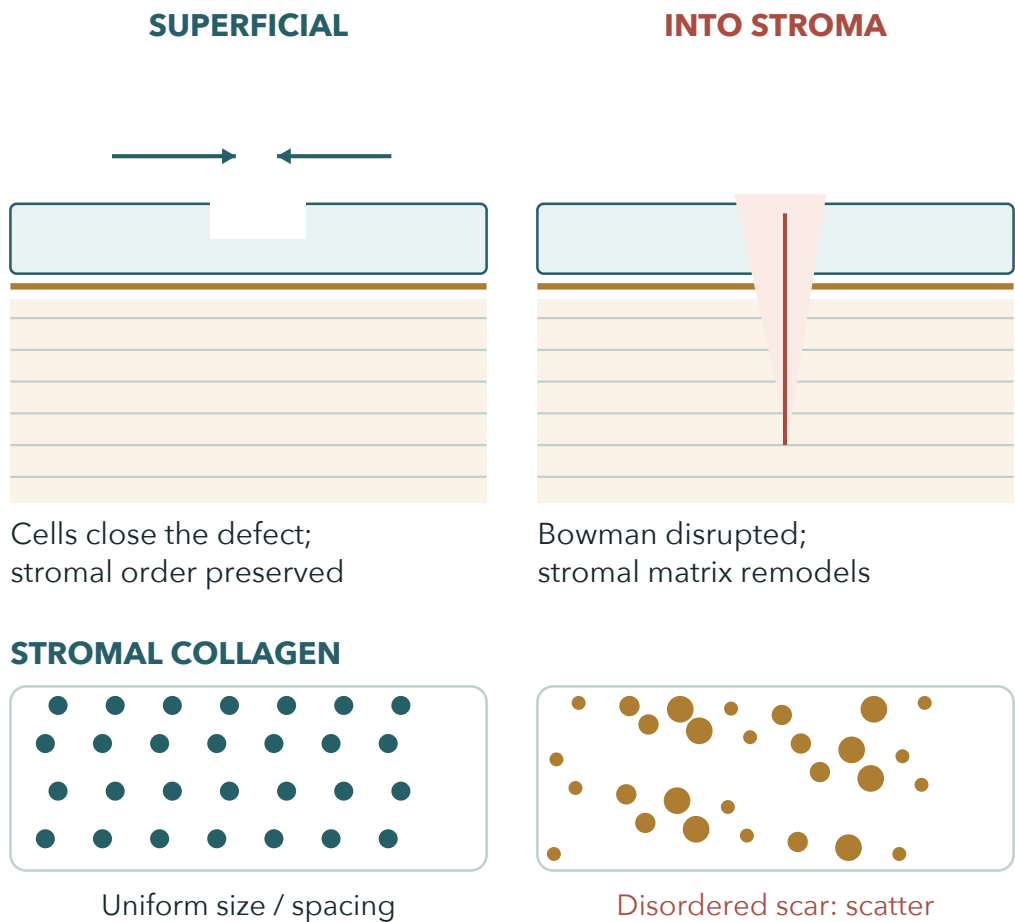
| Component              | Role                                                             |
|------------------------|------------------------------------------------------------------|
| Surface-cell junctions | Restrict passage between adjacent epithelial cells.              |
| Basement membrane      | Supports attachment and regulates epithelial-stromal signalling. |
| Limbal progenitors     | Supply renewing corneal epithelial cells.                        |

### Use the dye logically

Fluorescein highlights an epithelial defect. Loss of staining after closure indicates surface repair; it does not prove that a deeper stromal scar has cleared.

# Surface closure versus optical recovery

## The depth switch



A wound entering stroma may heal with disordered matrix. The illustrated collagen fields show optical order, not exact fibril dimensions.

## Depth changes the outcome

| Injury                              | Expected reasoning                                                       |
|-------------------------------------|--------------------------------------------------------------------------|
| Epithelium only                     | Regeneration can restore a clear surface over an intact scaffold.        |
| Bowman + anterior stroma            | Bowman does not normally regenerate; stromal fibrosis can scatter light. |
| Persistent basement-membrane damage | Profibrotic signalling can sustain myofibroblasts and haze.              |

## Why normal cornea transmits light

Uniform collagen fibril diameter, controlled spacing and avascularity limit scatter. Endothelial hydration control is another requirement, not a substitute for stromal order.

Sources NIH: Corneal repair | Corneal basement membranes | Corneal collagen structure